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(54) **ACCESS CONTROL CLASSIFIER TRAINING**

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(57) **ABSTRACT**

(65) **Prior Publication Data**

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A computer implemented method of access control for a user device having at least one component for determining behaviors of the user. The method including defining a training period during which access to the device is determined based on a credential-based authentication scheme wherein each access determination is used to generate an item of training data and training a machine learning classifier based on the training data such that the classifier is operable to classify user behavior as compliant or non-compliant. The method further including, in response to a determination that a behavior subsequent to the training period is classified as non-compliant, requesting a credential-based authentication of the user and permitting access to the device in response to the credential-based authentication, wherein permitting access to the device further includes constructively training the classifier based on the subsequent behavior as a compliant behavior.

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G06F 21/31 (2013.01)

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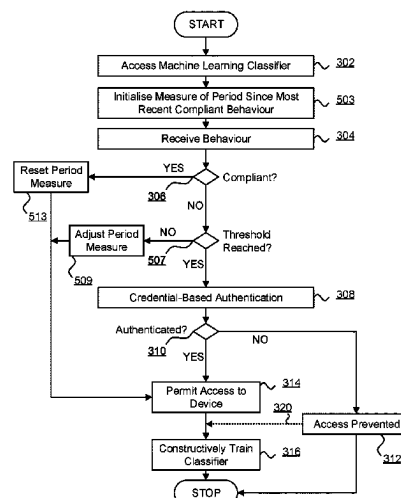
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CPC G06F 21/316; G06F 21/10; G06N 20/00

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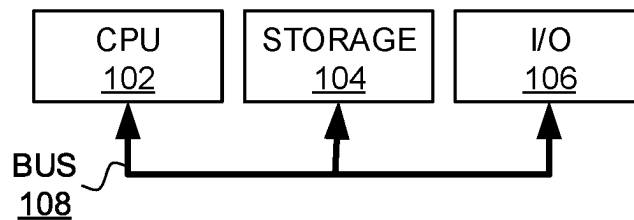
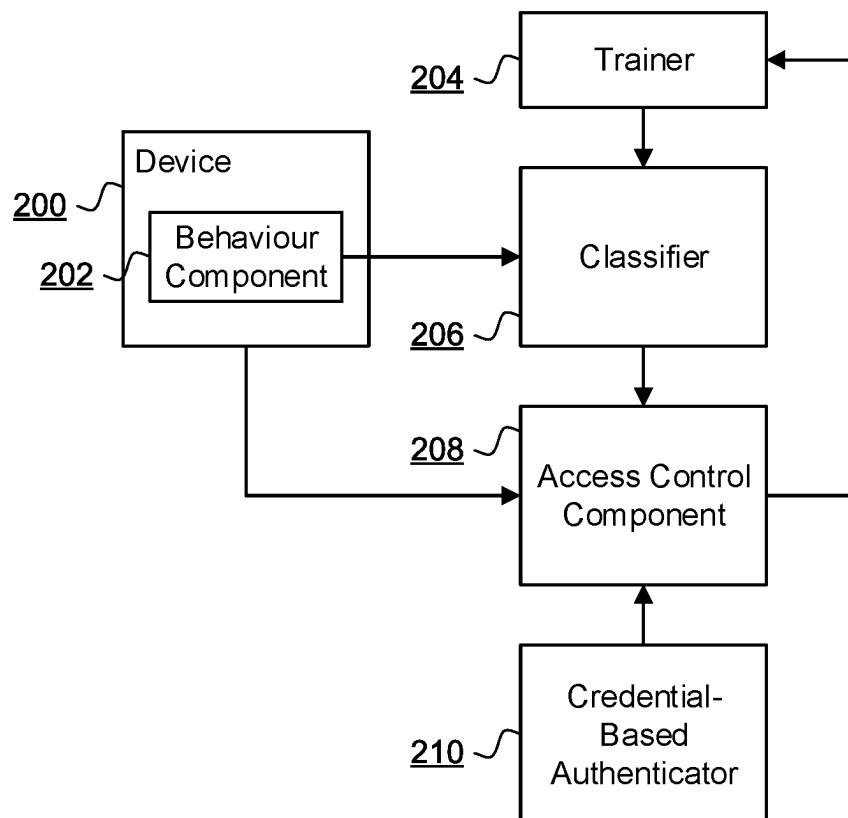
FIGURE 1**FIGURE 2**

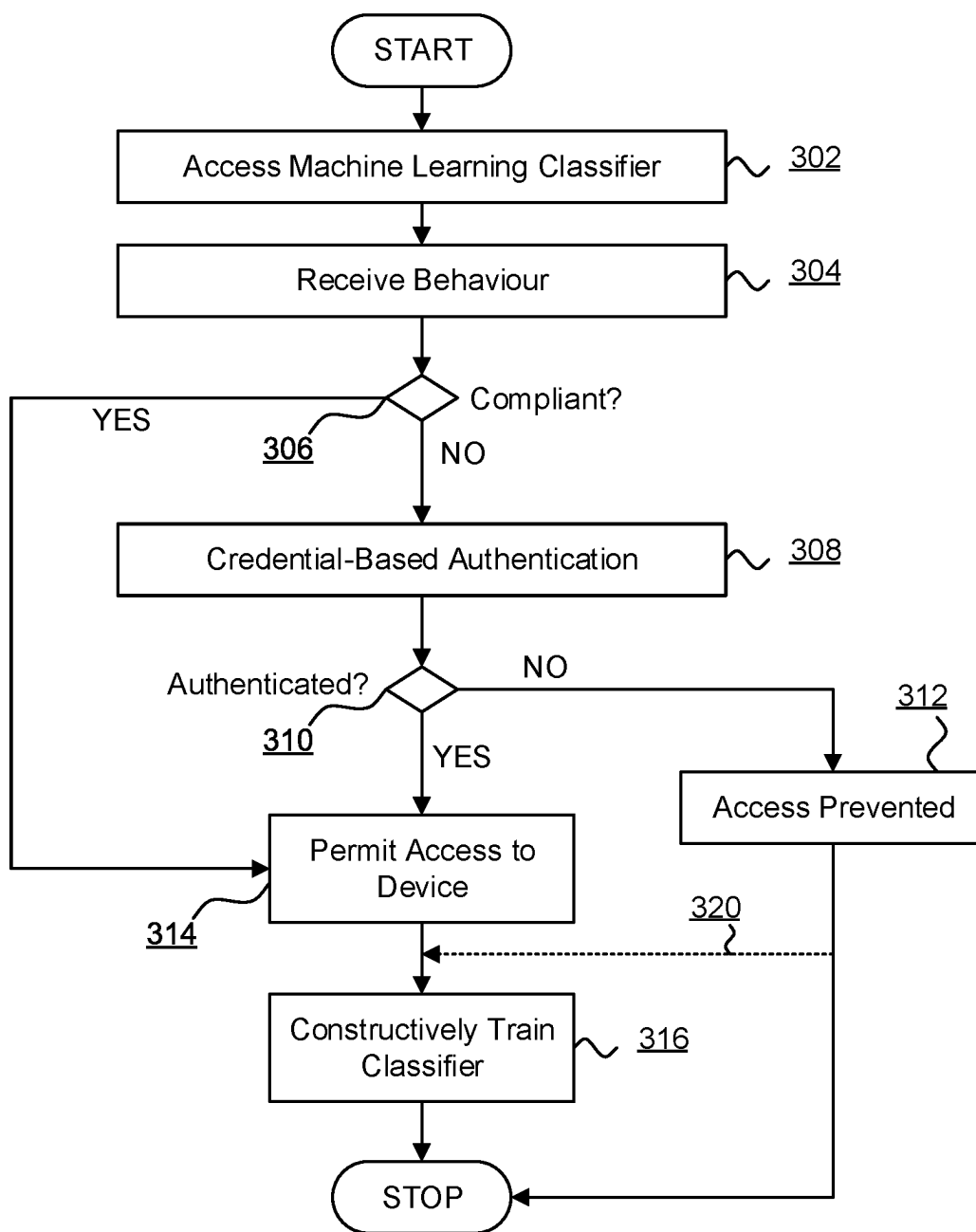
FIGURE 3

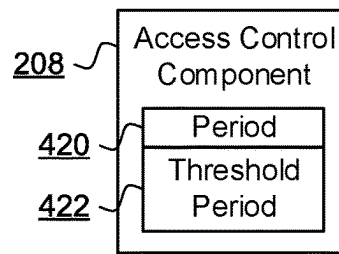
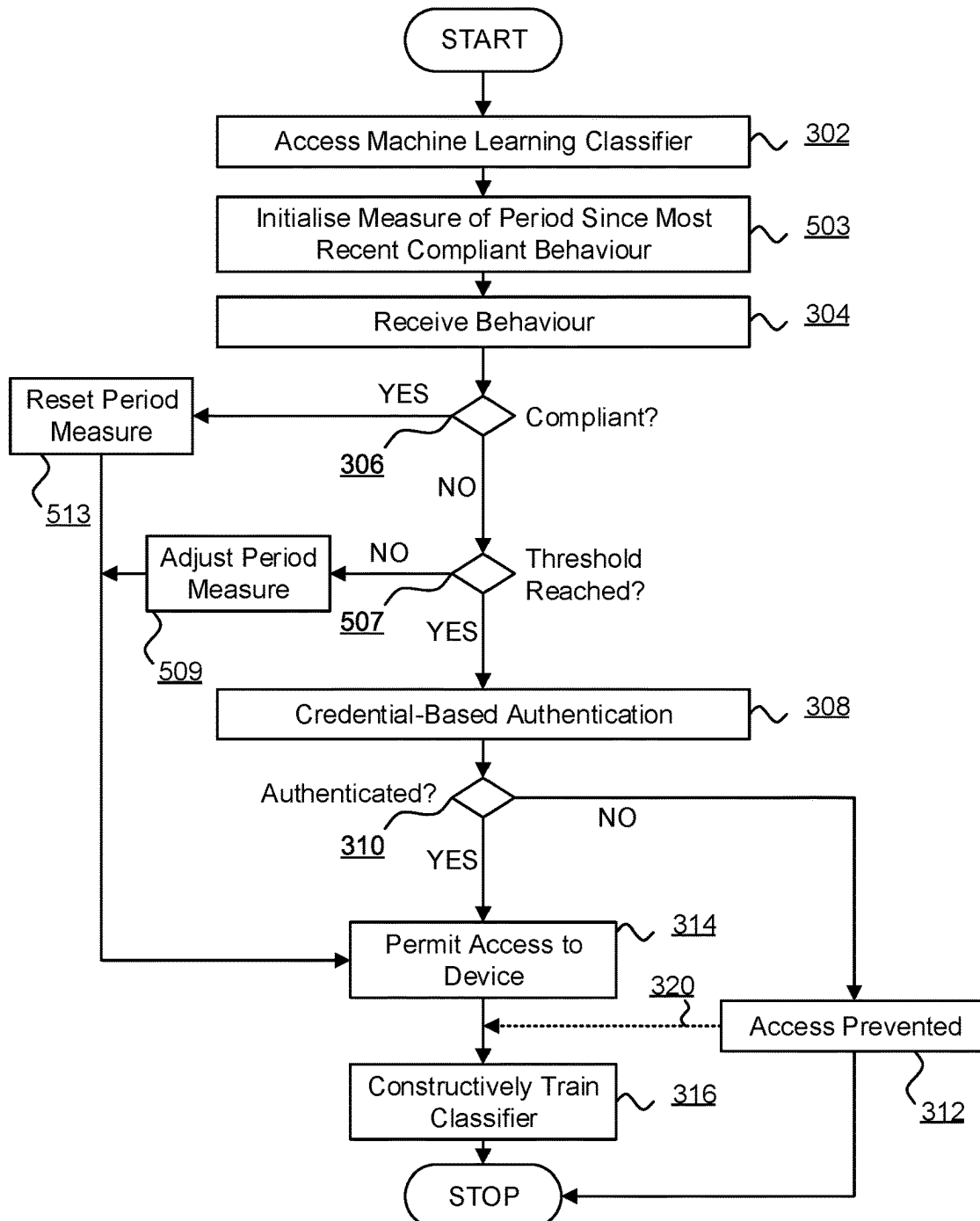
FIGURE 4**FIGURE 5**

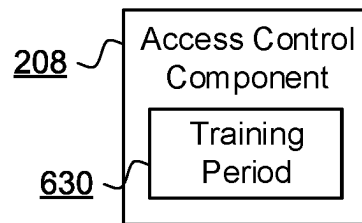
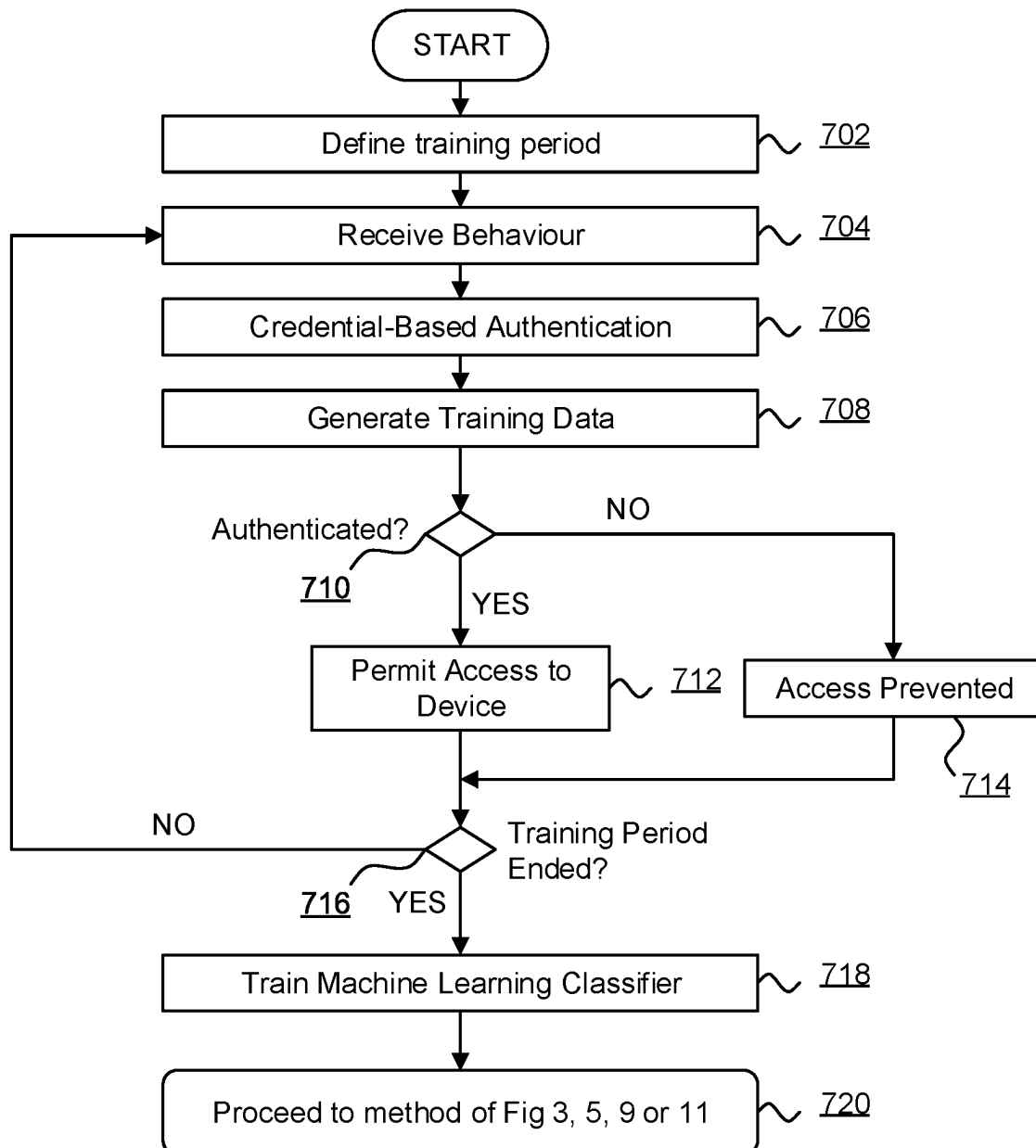
FIGURE 6**FIGURE 7**

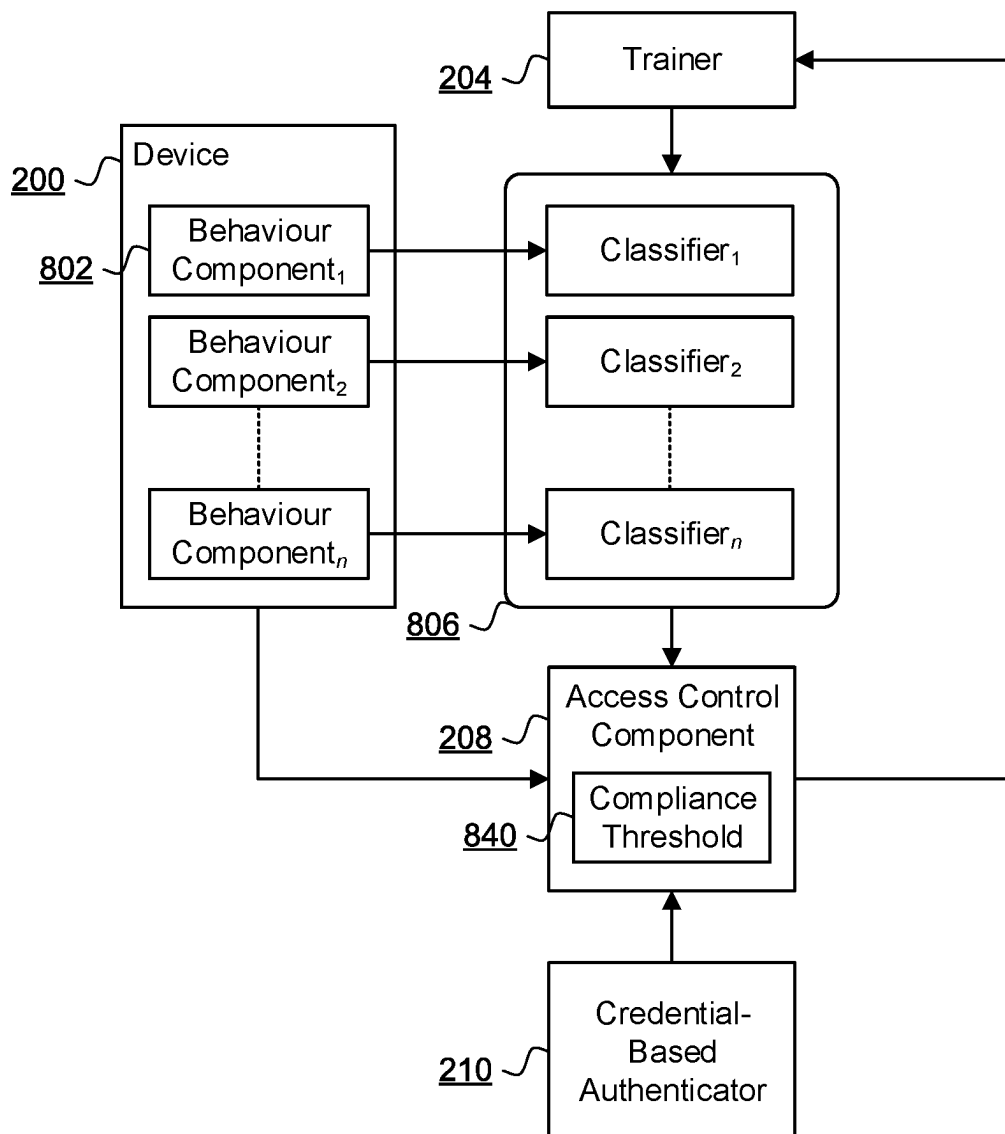
FIGURE 8

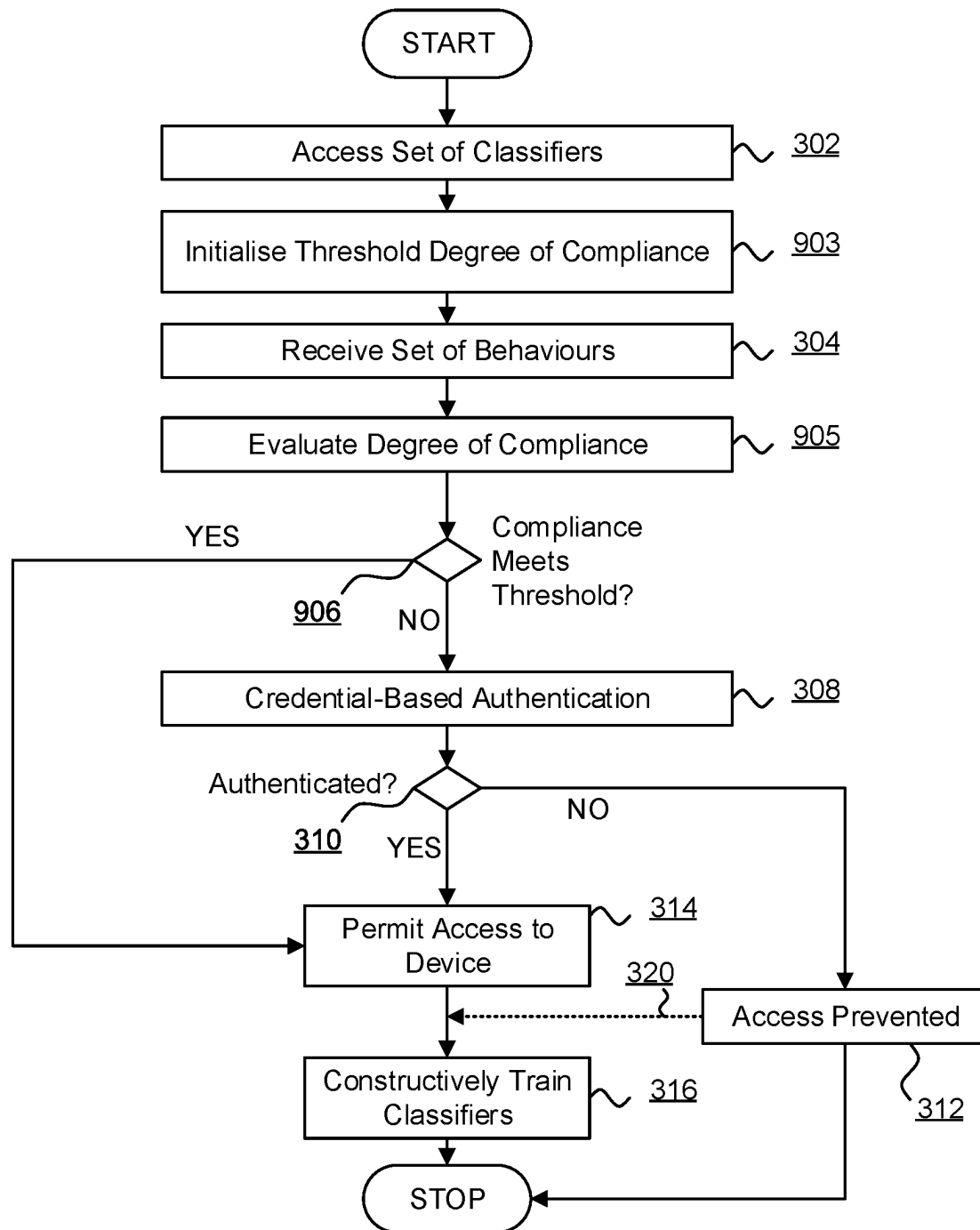
FIGURE 9

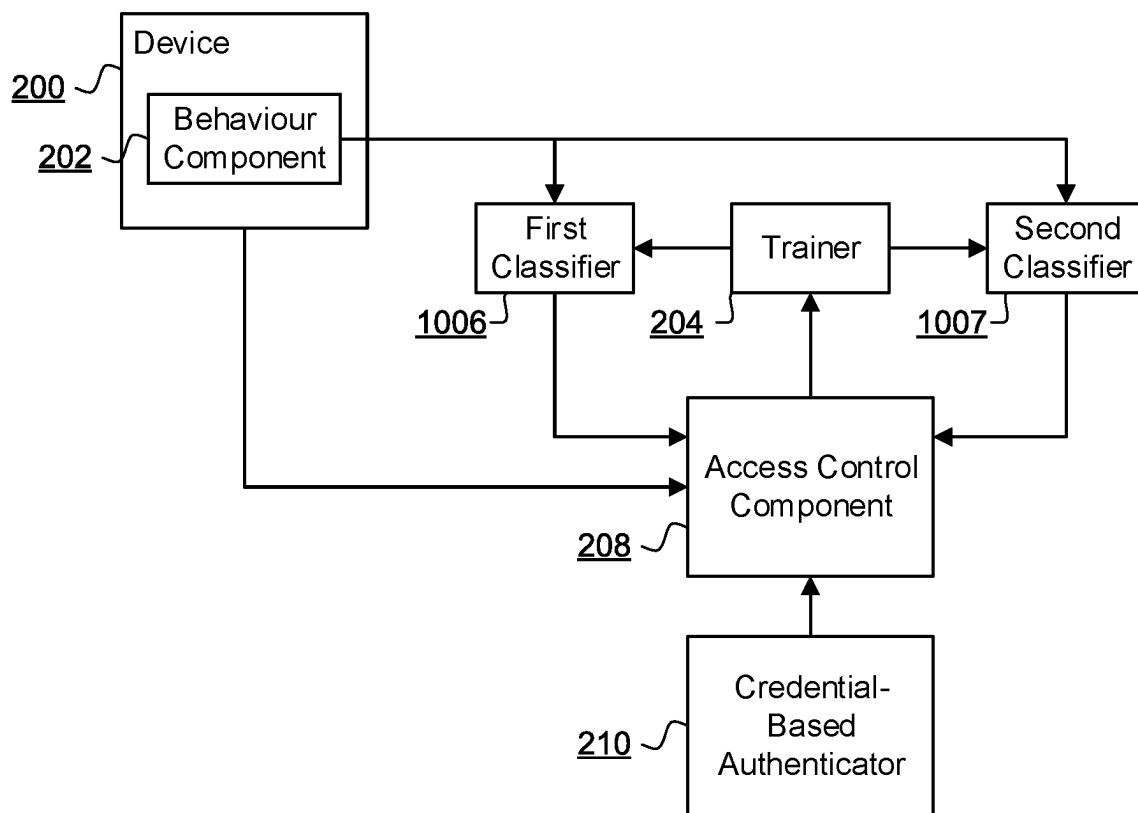
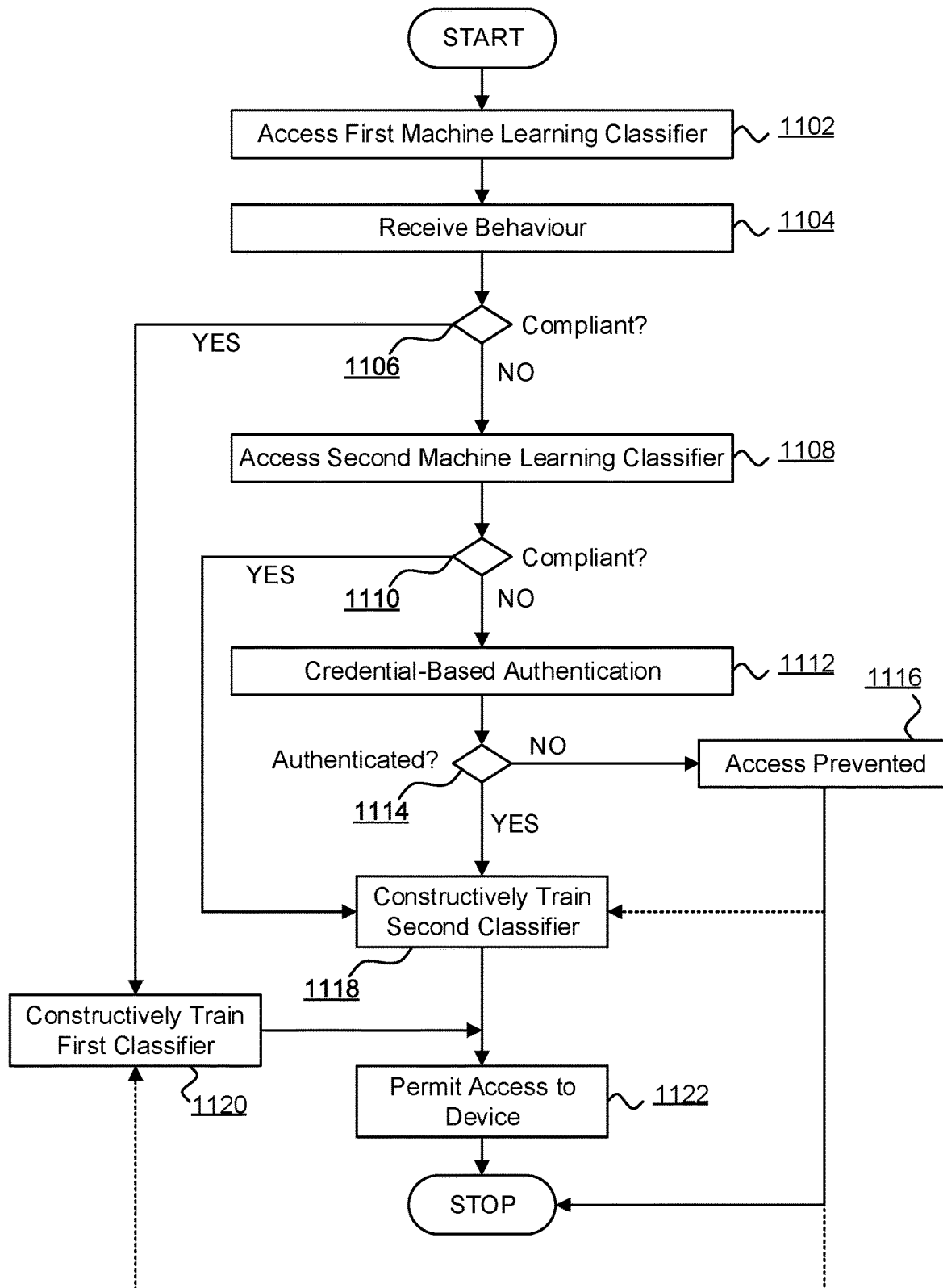
FIGURE 10

FIGURE 11

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ACCESS CONTROL CLASSIFIER TRAINING**PRIORITY CLAIM**

The present application is a National Phase entry of PCT Application No. PCT/EP2020/055372, filed Feb. 28, 2020, which claims priority from EP Patent Application No. 19161166.4, filed Mar. 7, 2019, each of which is hereby fully incorporated herein by reference.

FIELD

The present disclosure relates generally to access control for a user device.

BACKGROUND

User devices such as portable computing devices, tablet computers, smartphones, mobile terminals and the like are increasingly mobile and pervasive. Access control security measures for such devices typically involve a password, codeword, numeric or pattern-based lock or employ biometrics such as fingerprint or facial recognition. However, once access is gained to a device, the access is typically persistent while the device is in-use or at least until some expiry condition. Accordingly, accessing an unlocked device provides access to the device limited recourse to access control mechanisms.

Thus, there is a challenge in providing access control for user devices that address and/or mitigate these challenges.

SUMMARY

According to a first aspect of the present disclosure, there is provided a computer implemented method of access control for a user device having at least one component for determining behaviors of the user, the method comprising: defining a training period during which access to the device is determined based on a credential-based authentication scheme wherein each access determination is used to generate an item of training data including at least a result of the credential-based authentication and a behavior of the user; training a machine learning classifier based on the training data such that the classifier is operable to classify user behavior as compliant or non-compliant such that compliant behavior is determined by the classifier to be consistent with prior behavior for permitting access to the device subsequent to the training period; and responsive to a determination that a behavior subsequent to the training period is classified as non-compliant, requesting a credential-based authentication of the user and permitting access to the device in response to the credential-based authentication, wherein permitting access to the device further includes constructively training the classifier based on the subsequent behavior as a compliant behavior by providing the subsequent behavior as an additional training example.

In embodiments, preventing access to the device further includes constructively training the classifier based on the subsequent behavior as a non-compliant behavior by providing the subsequent behavior as an additional training example.

In embodiments, the component is one or more of: a location sensor; a position sensor; an orientation sensor; an accelerometer; an input device; a touch-screen; a temperature sensor; a time determiner; a pressure sensor; an olfactory sensor; a chemical sensor; a biometric sensor; a heart rate sensor; a cardiogram generator; a sound sensor; a voice

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recognition component; a handwriting recognition component; a global positioning system; and a gyroscope.

In embodiments, the credential-based authentication includes one or more of: an authentication scheme using a user identifier and password; a key-based user authentication scheme; a token-based user authentication scheme; and a multi-factor authentication scheme in which authentication is requested via a different device.

According to a second aspect of the present disclosure, there is provided a computer system including a processor and memory storing computer program code for performing the steps of the method set out above.

According to a third aspect of the present disclosure, there is provided a computer system including a processor and memory storing computer program code for performing the steps of the method set out above.

BRIEF DESCRIPTION OF DRAWINGS

Embodiments of the present disclosure will now be described, by way of example only, with reference to the accompanying drawings, in which:

FIG. 1 is a block diagram a computer system suitable for the operation of embodiments of the present disclosure.

FIG. 2 is a component diagram of an arrangement for access control for a user device according to an embodiment of the present disclosure.

FIG. 3 is a flowchart of a method of access control according to an embodiment of the present disclosure.

FIG. 4 is an alternative arrangement of the access control component of FIG. 2 in accordance with an embodiment of the present disclosure.

FIG. 5 is a flowchart of a permissive access control mechanism for a user device according to an embodiment of the present disclosure.

FIG. 6 is an alternative arrangement of the access control component of FIG. 2 in accordance with an embodiment of the present disclosure.

FIG. 7 is a flowchart of a method of training a classifier for use with the embodiments of FIGS. 3, 5, 9 and/or 11 in accordance with an embodiment of the present disclosure.

FIG. 8 is a component diagram of an arrangement for behavior-based access control for a user device according to an embodiment of the present disclosure.

FIG. 9 is a flowchart of a method of access control for the embodiment of FIG. 8 in accordance with an embodiment of the present disclosure.

FIG. 10 is a component diagram of an arrangement for multi-classifier access control for a user device according to an embodiment of the present disclosure.

FIG. 11 is a flowchart of a method of access control for the embodiment of FIG. 10 in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

FIG. 1 is a block diagram of a computer system suitable for the operation of embodiments of the present disclosure. A central processor unit (CPU) 102 is communicatively connected to a storage 104 and an input/output (I/O) interface 106 via a data bus 108. The storage 104 can be any read/write storage device such as a random-access memory (RAM) or a non-volatile storage device. An example of a non-volatile storage device includes a disk or tape storage device. The I/O interface 106 is an interface to devices for the input or output of data, or for both input and output of data. Examples of I/O devices connectable to I/O interface

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106 include a keyboard, a mouse, a display (such as a monitor) and a network connection.

FIG. 2 is a component diagram of an arrangement for access control for a user device 200 according to an embodiment of the present disclosure. The user device 200 is a computing device such as a portable computing device, tablet computer, smartphone, mobile terminal, gaming device, in-vehicle information system, in-vehicle control device, medical device, communications device, entertainment device or other device as will be apparent to those skilled in the art and as may be conceived even after the relevant date of the present application. The user device 200 is used by at least one user in a manner that constitutes user behavior. Such behaviors can include, by way of example only and inter alia: a particular holding, handling, orientation and/or movement of the device 200; a rotation, degree of pressure, speed of interaction, frequency of interaction of the device 200; a location, position, relative location, speed, velocity or altitude of the device; an environmental characteristic, temperature, humidity or pressure of the device; the use of a particular hand, finger, limp, facial feature to handle, hold or interact with the device; particular biometric characteristics of the user using the device; characteristics of use of the device including particular user interaction behaviors such as user interface gestures, characteristic interactions, swipes, button presses, style of interaction, pressure of interaction; and other behavioral characteristics as will be apparent to those skilled in the art. Notably, certain of the exemplary behaviors are described in terms characteristics of a user, an environment, the device or interaction therebetween and these are considered to constitute behaviors of the user who, for example, places the device in such environment, or interacts using such user characteristics.

The device 200 includes at least one behavior component 202 as a hardware, software, firmware or combination component adapted to determine a behavior of a user of the device 200. For example, the behavior component 202 can include one or more of, inter alia: a location sensor; a position sensor; an orientation sensor; an accelerometer; an input device; a touch-screen; a temperature sensor; a time determiner; a pressure sensor; an olfactory sensor; a chemical sensor; a biometric sensor; a heart rate sensor; a cardiogram generator; a sound sensor; a voice recognition component; a handwriting recognition component; a global positioning system; a gyroscope; and other behavior components as will be apparent to those skilled in the art.

An access control component 208 is provided as a hardware, software, firmware or combination component for providing access control for the device 200. Access control includes determining, for a user seeking to access or accessing the device 200, whether the user's access is authorized and providing or continuing access for such user, or alternatively whether the user's access is not authorized and preventing access for such user. Preventing access can include, for example, locking, powering off, disabling or otherwise securing the device 200 from use by the unauthorized user.

The access control component 208 is arranged to perform access control interventions for the device 200 by which access to the device 200 by a user is granted or prevented, the interventions being triggered according to one or more predetermined rules. In one embodiment, the access control component 208 is configured to undertake continuous authentication of a user such that access control is implemented on a regular, routine or substantially continuous basis, such as, for example, inter alia: every predetermined period of time according to a particular frequency; whenever

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a particular operation or class of operation is performed by the device 200 such as opening an application, performing a data transfer, accessing a screen or the like; on the basis of a stochastic variable such that authentication is triggered periodically stochastically; or on some other continuous, regular or similar basis as will be apparent to those skilled in the art.

The access control component 208 is operable with a classifier 206 constituted as a machine learning algorithm such as, by way of example: a perceptron classifier; a naive Bayes classifier; a decision tree classifier; a logistic regression classifier; a k-nearest neighbor classifier; an artificial neural network or deep learning classifier; a support vector machine or other classifier as will be apparent to those skilled in the art. The classifier is trained by a trainer 204 as a software, hardware, firmware or combination component adapted to undertake training of the classifier 206 based on training examples. In particular, the classifier 206 is trained based on behavior of a user (or, conceivably, multiple users) of the device 200 determined by the behavior component 202 such that the classifier is able to classify subsequent user behavior as either compliant with training examples for an authorized user or non-compliant. Notably, a classification of non-compliance is a classification that behavior is not consistent with behavior of an authorized user as learned by the classifier 206. In some embodiments, the classifier 206 can be further trained with training examples including behavior of non-authorized users such that the determination of non-compliance by way of the classifier 206 can also be effected as a consistency of behavior with the training examples for non-authorized users. Accordingly, the trainer 204 initially trains the classifier 206 with at least one prior behavior of an authorized user, each training example including a characterization of a behavior and, in some embodiments, an indication of whether such behavior related to an authorized or un-authorized user. Behavior characteristics can be encoded for use in training and sampling the classifier 206 using any suitable method such as a one-hot or multi-hot vector encoding.

Thus, in use, the access control component 208 provides a determination of whether a user of the device 200 is authorized to access the device based on behavior characteristics received from the behavior component 202 and with reference to the classifier 206. According to embodiments of the present disclosure, the access control component 208 is further adapted to trigger or perform additional training of the classifier 206 responsive to at least positive authentications of a user accessing the device 200 so as to further improve a model, data structure, neural network or other machine learning construct of the classifier 206 and, therefore, the suitability of the classifier 206 for classifying user behaviors as compliant (for authorization) or non-compliant (for non-authorization). Thus, on determination of at least a positive authorization of a user based on a behavior classification for the user, the access control component 208 formulates a new training example including the user's behavior and, depending on the machine learning approach, its classification as a compliant behavior, and triggers the trainer 204 to perform a further training process of the classifier using such new training example, such further training being additional to all prior training of the classifier 206. Thus, the classifier 206 is constructively trained in that it is trained additionally to its existing training so as to progressively improve the suitability of the classifier 206.

Where user behavior is classified by the classifier 206 as non-compliant, the access control component 208 is arranged to invoke a credential-based authenticator 210 to

confirm a state of authorization for the user. The credential-based authenticator **210** is a hardware, software, firmware or combination component for authenticating a user's authorization to access the device **200** based on credential and/or challenge responses of the user. For example, a credential-based authentication scheme can include, inter alia: an authentication scheme using a user identifier and password; a key-based user authentication scheme; a token-based user authentication scheme; a multi-factor authentication scheme in which authentication is requested via a different device, and other credential-based authentication schemes as will be apparent to those skilled in the art. The premise of a credential-based authentication scheme is that an ability to present authorized credentials or suitably respond to an authentication challenge serves to confirm a state of authorization to access the device **200**. Thus, the credential-based authentication scheme is used as a definitive confirmation of a state of authorization of a user. Thus, a user exhibiting behavior classified by the classifier **206** as non-compliant but nonetheless able to successfully authenticate using the credential-based authentication scheme is permitted to access the device **200** by the access control component **208**. Furthermore, such a user exhibiting behavior classified as non-compliant and nonetheless being permitted access to the device **200** based on the credential-based authentication scheme is used by the access control component **208** to generate a new training example for the classifier **206**. Such new training example is used to constructively train the classifier **206** in the manner previously described so as to improve the suitability of the classifier to classify behavior such as that which was previously classified as non-compliant but which was nonetheless determined to be behavior of an authorized user based on the credential-based authentication scheme. Notably, the new training example in the case of a user authenticated by a credential-based authentication scheme, will include the behavior of the user and the credential-based authentication outcome: i.e., that the user is authorized and, thus, should be deemed compliant in the classifier.

Where user behavior is classified by the classifier **206** as non-compliant and the user fails to successfully authenticate using the credential-based authentication scheme, access to the device for the user is prevented. In this way, unauthorized users are prevented from accessing the device **200**. In embodiments, the access control component **208** is further adapted to trigger or perform additional training of the classifier **206** responsive also to negative authentications of a user accessing the device **200** so as to further improve a model, data structure, neural network or other machine learning construct of the classifier **206** and, therefore, the suitability of the classifier **206** for classifying user behaviors as compliant (for authorization) or non-compliant (for non-authorization). That is, where a user is found to be non-compliant based on the classifier **206** and the user fails to authenticate using the credential-based authentication scheme, the behavior of such user can be used to reinforce the non-compliance in the classifier **206** by a further training example generated from the behavior and the non-authorized status of the user.

In this way, the classifier **206** is progressively improved through constructive training during repeated authentication processes.

It will be appreciated that, while the access control component **208**, the classifier **206**, the trainer **204** and the credential-based authenticator **210** are illustrated in FIG. 2 (and other Figures) as being separate from the device **200**, any or all of these components can be included in, integrated

with, provided by, stored in and/or executed by the user device **200**. In an embodiment, all of these components are provided as services, facilities or features of the device **200** such that the device **200** is operable to perform access control without external assistance. In alternative embodiments, any of the components **204** to **210** may be provided by a separate computer system such as a network connected computer system accessible to the device **200** by way of a communications network, such as by secure network communication using encryption. Furthermore, any of components **204** to **210** can be provided as cloud based or virtual services. The same applied to the other embodiments described with respect to FIGS. 4 to 11.

FIG. 3 is a flowchart of a method of access control according to an embodiment of the present disclosure. Initially, at **302**, the access control component **208** accesses the machine learning classifier **206** that is trained based on at least one prior behavior of a user using the device. At **304** the access control component **208** receives a subsequent behavior of a user and determines, at **306**, if the subsequent behavior is compliant by classifying the subsequent behavior using the classifier **206**. If the classifier **206** classifies the behavior as compliant the method proceeds to **314**. Otherwise, the method proceeds to **308**. At **308** the access control component **208** employs a credential-based authentication scheme by way of the credential-based authenticator **210** to authenticate the user. If the user fails to authenticate by the credential-based authentication at **310** the method proceeds to **312** where access to the device is prevented or ceased. Alternatively, if the method determines, at **310**, that the user successfully authenticates by the credential-based authentication, the method proceeds to **314**.

314 is arrived at either from **306** in the event of a compliant classification of user behavior, or from **310** in the event of successful authentication of the user by the credential-based authentication scheme. At **314** the access control component **208** permits the user access to the device **200**. At **316**, the access control component **208** generates a new training example for the trainer **204** to constructively train the classifier **206**, the training example being based on the subsequent user behavior and the appropriate authentication outcome. Notably, the appropriate authentication outcome for users authorized by way of compliant classification or authenticated by way of the credential-based authentication scheme is "authorized" (or "compliant"), even if (in the event of credential-based authorized users) the original classification for the user was non-compliant.

As previously described, in the event of non-compliant behavior of a user that further fails to authenticate by way of the credential-based authentication scheme, access is prevented as **312**. In some embodiments, the behavior, non-compliant and non-authorized state of such a user is additionally used as a new training example at **316** to constructively train the classifier **206**. This serves to improve the classifier's suitability in respect of non-authorized users. This optional operation from **312** is indicated by a broken line in FIG. 3.

FIG. 4 is an alternative arrangement of the access control component **208** of FIG. 2 in accordance with an embodiment of the present disclosure. Notably, the access control component **208** of FIG. 4 is identical to, and used in the same arrangement as, that described with respect to FIG. 2 except for differences described below.

The access control component **208** includes a period **420** as a variable, parameter or other suitable data item for storing an indication of a period since a most recent compliant behavior of a user occurred. Notably, the period can

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be a time period measured in, for example, seconds, minutes and/or hours. Thus, the period can be recorded as a time of a most recent compliant behavior such that a period since such time can be readily evaluated. Alternatively, the period can be a measure of a number of authentication attempts that have occurred resulting in non-compliance. For example, the period can be a measure of a number of occasions when access to the device **200** is prevented based on non-compliant behavior.

The access control component **208** further includes a threshold period **422** as a predefined size, length or value of a period (whether measured in terms of, for example, time or non-compliant behavior classifications), beyond which access to the device **200** is to be prevented. In use, the access control component **208** of FIG. **4** operates in substantially the same way as that of FIG. **2** except that non-compliant behavior (as determined by the classifier **206**) does not lead to credential-based authentication and potential access prevention until a period **420** since a most recent compliant behavior meets or exceeds the threshold period **422**. Thus, in this way, the access control component **208** can be seen to be permissive in the event of non-compliant behavior for a predetermined period up to the threshold period **422**. Such a permissive approach to authentication is useful, for example, for non-critical devices, services or resources such as subscription resources where non-compliance indicates non-subscription (e.g. a user has not paid to use the device **200**) but cessation of access can be postponed for a period (the threshold period **422**) in the event that behavior transitions to compliant behavior. A tangible example of the usefulness of such an approach can be seen in the event that an authorized user of the device **200** (e.g. a paid-up subscriber) hands the device to a third party briefly for demonstration purposes. The third party is not a paid-up subscriber and is not an authorized user. The third party behaviors will be classified as non-compliant, but for the period up to the threshold period **422** access is not prevented such that the third party can utilize the device within the period. Once the threshold period is reached then the access control component **208** reverts to the approach described with respect to FIG. **2** in which a credential-based authentication scheme is invoked.

FIG. **5** is a flowchart of a permissive access control mechanism for a user device **200** according to an embodiment of the present disclosure. Many of the operations of FIG. **5** are identical to those described above with respect to FIG. **3** and a description of these will not be repeated here. Where FIG. **5** differs from FIG. **3**, FIG. **5** additionally includes **503** following **302** at which the measure of a period **420** since a most recent compliant behavior is initialized, for example, by setting the period **420** to zero or, in the case of a period measured in time, setting the period **420** to a current or predetermined time. Further, in FIG. **5**, where a user behavior is classified as compliant at **306**, the method proceeds to reset the period since most recent compliant behavior at **513**, for example by setting the period **420** to zero or, in the case of a period measured in time, setting the period **420** to a current time. In FIG. **5**, where a user behavior is classified as non-compliant at **306**, the method proceeds to **507** where the access control component **208** determines if the threshold period **422** is reached. This can be achieved by a comparison between the period **420** and the threshold **422**. While the threshold is not reached the method proceeds to **509** where the period measure **420** is adjusted by, for example, incrementing it or updating it to indicate an increase in the time since a most recent compliant behavior. Subsequently, **509** leads to **314** where access to the device

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is permitted while the threshold is not reached. Where the threshold period is determined to be reached at **507**, the method proceeds to **308** where credential-based authentication is performed as described above with respect to FIG. **3**. All other operations of FIG. **5** are as previously described with respect to FIG. **3**.

FIG. **6** is an alternative arrangement of the access control component **208** of FIG. **2** in accordance with an embodiment of the present disclosure. Notably, the access control component **208** of FIG. **6** is identical to, and used in the same arrangement as, that described previously with respect to FIG. **2** and/or FIG. **4** except for differences described below.

The access control component **208** of FIG. **6** is adapted for initial training of the classifier **206** such that the classifier **206** is suitable for classifying behavior of a user as compliant or non-compliant as indicative of authorized and non-authorized users respectively. Notable, access control component **208** as will be described here with reference to FIGS. **6** and **7** is used to undertake or trigger initial training of the classifier **206**. This initial training is required for a classifier **206** that is as yet untrained or not sufficiently trained for the purpose of use in the arrangements of, for example, FIGS. **2**, **3**, **4** and **5**. Thus, the training triggered or performed by the access control component **208** during the authentication process of, for example, FIGS. **3** and **5**, is the same as previously described. The initial training provided by the access control component **208** here and with respect to FIGS. **6** and **7** is additional to this.

The access control component **208** of FIG. **6** further includes a training period **630** variable, parameter or other suitable data item. The training period **630** is a predetermined indication of a period during which training of the classifier **206** is performed and during which all authentication of users to access the device **200** is performed using the credential-based authenticator **210** as previously described. The training period **630** can be constituted as, for example, a time period, a duration, a number of training examples, a number of authentication attempts or other suitable definition of a period for training as will be apparent to those skilled in the art. In use, the access control component **208** determines if an authentication is to take place within the training period **630** and, if so, performs a credential-based authentication process to authenticate a user. Behavior information is nonetheless received by the access control component **208** and is used along with the result of the credential-based authentication process to generate training data as training examples for training the classifier **206** by the trainer **204**. Notably, training examples are generated for both authenticated users (such examples including the user behavior and compliant state), and non-authenticated users (such examples including the user behavior and non-compliant state). Once the training period **630** has ended, passed or otherwise expired then the training examples are used to train the classifier **206** such that the classifier **206** is suitable for use in methods of behavioral authentication such as those described with respect to FIG. **3**, **5**, **9** or **11**.

FIG. **7** is a flowchart of a method of training a classifier **206** for use with the embodiments of FIGS. **3**, **5**, **9** and/or **11** in accordance with an embodiment of the present disclosure. Initially, at **702**, the training period **630** is defined. At **704** user behavior for a user is received from the behavior component **202**. At **706** a credential-based authentication scheme is used to determine a state of authentication of the user—either authorized or non-authorized. At **708**, a new training example is generated as training data for the classifier **206** based on the received behavior information and the state of authorization for the user. At **710**, **712** and **714**

access to the device is permitted or prevented in dependence on the state of authentication of the user based on the credential-based authentication scheme. At **716** the method determines if the training period has ended and loops to **704** until the training period has ended. Once the training period ends the method proceeds to **718** where the access control component **208** undertakes or triggers training of the classifier **206** by the trainer **204** based on the training examples generated at the iterated **708**. Subsequently, the trained classifier **206** is used in methods of behavioral authentication such as those described with respect to FIG. 3, 5, 9 or 11.

FIG. 8 is a component diagram of an arrangement for behavior-based access control for a user device **200** according to an embodiment of the present disclosure. Many of the elements of FIG. 8 are identical to those described above with respect to FIG. 2 and these will not be repeated here. The device **200** of FIG. 8 includes a plurality of behavior components **802** each adapted to provide different behavior information for a user of the device. For example, each behavior component **802** can constitute a different sensor or set of sensors for determining behavior of a user. Further, in FIG. 8 multiple classifiers **806** are provided, each corresponding to a behavior component **802** of the device. While each of the classifiers **806** acts substantially as previously described with respect to FIGS. 1 to 7, the multiplicity of classifiers **806** is such as to provide a classifier for each different behavior information received from the different multiple behavior components **802**. Thus, for example: a location sensor behavior component can have associated a classifier for classifying location information; a user gesture behavior component can have associated a classifier for classifying user gestures; and so on.

Further, the access control component **208** of FIG. 8 includes a predetermined compliance threshold **840** as a parameter, variable or other data item for determining a strength, confidence or other degree of compliance of all classifiers **806** in aggregate required in order to authenticate a user and provide access to the device **200**. Thus, in use, each classifier **806** classifies behavior information from the behavior components **802** as previously described in order to determine a classification as either compliant or non-compliant. Furthermore, each classifier is operable to provide a strength, confidence or other degree of membership of received behavior with its determined classification such that a degree of membership of behavior with a compliant or non-compliant classification is provided. Such a degree of membership can be provided as, for example, a weight of connections in a neural network leading to the classification outcome, a confidence measure of the classifier, or other suitable measure of a degree of membership of behavior with a classification. For example, state vector machines can allow the retrieval of classification confidence on which basis a prediction classification is generated. Alternatively, a classifier **806** that provides a probabilistic score for its classifications, such as logistic regression classifiers, can be used whereby the score can constitute such degree of membership.

The degree of membership of each classifier for behaviors of a user are aggregated such as by statistical average, summation, or other suitable means. Thus, a degree of compliance of a user is evaluated. This degree of compliance is compared with the compliance threshold **840** and only where the threshold degree is met will compliance of a user be determined for the purpose of permitting access to the device. Non-compliance can result in the use of a credential-based authentication scheme as previously described.

FIG. 9 is a flowchart of a method of access control for the embodiment of FIG. 8 in accordance with an embodiment of the present disclosure. Many of the operations of FIG. 9 are identical to those described previously with respect to FIG. 2 and these will not be repeated here. FIG. 9 additionally includes, after **302**, new operation **903** in which the threshold degree of compliance **840** is initialized to a predetermined threshold degree. Subsequently, after receipt of a set of behaviors **304** from the behavior components **802**, the method evaluates a degree of compliance at **905** on aggregate for all behaviors based on the classifiers **206** for the behaviors. At **906** the method determines if the degree of compliance meets the threshold degree **840** and, if the threshold is met, the method proceeds to **314** where access to the device is permitted. Alternatively, if the threshold is not met, the method proceeds to **308** where a credential-based authentication scheme is employed. All other operations of the method of FIG. 9 are identical to those described with respect to FIG. 2.

FIG. 10 is a component diagram of an arrangement for multi-classifier access control for a user device **200** according to an embodiment of the present disclosure. Many of the elements of FIG. 10 are identical to those described above with respect to FIGS. 2, 4, 6 and 8 and these will not be repeated here. FIG. 10 includes a pair of classifiers—a first classifier **1006** and a second classifier **1007**, each of which is substantially consistent with the classifiers as hereinbefore described. The first and second classifiers **1006**, **1007** of FIG. 10 are used to provide a multi-level classification scheme of user behavior such that atypical behavior that is classified as non-compliant by the first classifier **1006** can be processed by the second classifier **1007** to determine if such atypical behavior is nonetheless typical for the user of non-compliant behavior according to the first classifier **1006**. Such an approach is particularly beneficial where the device **200** transitions between two distinct contexts such as: a primary context in which the device is primarily used and in which user behavior is modeled by the first classifier **1006** as most typical or prevalent behavior; and a secondary context in which the device is occasionally used in which user behavior is modeled by the second classifier **1007**. Such different contexts are widely experienced in real use-cases, with examples including: a secondary context in which the device **200** is used in an aircraft or other vehicle that is occasional and can restrict, constrain or otherwise affect user behavior due to, for example, constraints of space or usage permission; a secondary context in which the device **200** is occasionally used in different environmental conditions such as in the dark, at night, in a sunny location or the like, such that user behavior is affected due to, for example, a need to operate the device differently to provide illumination or avoid glare or the like; a secondary context in which the device is put to a different use-case such as, for example, an everyday device being used as a presentation device for sharing content with a group in which case the device may be handled, operated or otherwise used differently exhibited by differing behaviors; and other such secondary contexts in which behaviors occasionally differ from a primary context as will be apparent to those skilled in the art. Notably, the exact nature, cause and type of such distinction between a primary and secondary context are not important to embodiments of the present disclosure which are arranged to learn classification schemes and model behavior in contexts irrespective of the nature or cause.

Thus, in the arrangement of FIG. 10, the first classifier **1006** operates substantially as previously described with respect to FIG. 2 to classify user behavior to determine

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compliance or non-compliance as a basis for access control for the device **200**. The second classifier **1007** provides a secondary classification scheme for behaviors that are classified non-compliant by the first classifier **1006** but where a user otherwise successfully authenticates using a credential-based authentication scheme. The second classifier **1007** is therefore trained based on training examples arising from behaviors found non-compliant by the first classifier **1006** so as to classify such behaviors as compliant or non-compliant with typical behaviors in a secondary user context. In embodiments, such behaviors found to be non-compliant by the first classifier **1006** but authenticated by the credential-based authenticator **210** are not used as training examples for the first classifier **1006** to provide a distance between the two classifiers **1006**, **1007** allowing each classifier to specialize in their respective user contexts. In contrast, behaviors leading to a complete failure to authenticate by classification as non-compliant by both the first and second classifiers **1006**, **1007** and a failure to authenticate by credential-based authentication can be used as training examples for both classifiers **1006**, **1007** since such behaviors are indicative of non-compliant user behavior in both primary and secondary user contexts and for which device access should be prevented.

FIG. **11** is a flowchart of a method of access control for the embodiment of FIG. **10** in accordance with an embodiment of the present disclosure. Initially, at **1102**, the method accesses the first machine learning classifier **1006** and receives a user behavior at **1104**. The first classifier **1006** classifies the received behavior and where the behavior is classified as compliant at **1106** the method proceeds to **1120** where the first classifier is constructively trained using the behavior and compliant status as a new training example. Following constructive training at **1120**, access to the device is provided to the user at **1122**.

Where the behavior is determined at **1106** to be non-compliant based on the first classifier **1006**, the method proceeds to **1108**. At **1108** the second classifier **1007** is accessed and the behavior is classified according to the second classifier **1007**. Where the behavior is classified as compliant by the second classifier **1007** at **1110** the method proceeds to **1118** where the second classifier **1007** is constructively trained using the behavior and compliant status as a new training example. Following constructive training at **1118**, access to the device is provided to the user at **1122**.

Where the behavior is determined at **1110** to be non-compliant based on the second classifier **1007**, the method proceeds to **1112** where a credential-based authentication scheme is employed to authenticate the user. If the user is successfully authenticated using the credential-based authentication scheme at **1114**, the method proceeds to **1118** which as been previously described. If the user is not successfully authenticated using the credential-based authentication scheme at **1114**, the method proceeds to **1116** where access to the device **200** is prevented. Notably, on failure to authenticate by the credential-based authentication scheme, the behavior and its status as non-compliant in both the first and second classifiers **1006**, **1007** can be used to constructively train either or both classifiers **1006**, **1007** to improve the suitability of the classifiers **1006**, **1007** to classify non-compliance effectively. This is indicated by the logical flows in FIG. **11** depicted by broken lines.

Insofar as embodiments of the disclosure described are implementable, at least in part, using a software-controlled programmable processing device, such as a microprocessor, digital signal processor or other processing device, data processing apparatus or system, it will be appreciated that a

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computer program for configuring a programmable device, apparatus or system to implement the foregoing described methods is envisaged as an aspect of the present disclosure. The computer program may be embodied as source code or undergo compilation for implementation on a processing device, apparatus or system or may be embodied as object code, for example.

Suitably, the computer program is stored on a carrier medium in machine or device readable form, for example in solid-state memory, magnetic memory such as disk or tape, optically or magneto-optically readable memory such as compact disk or digital versatile disk etc., and the processing device utilizes the program or a part thereof to configure it for operation. The computer program may be supplied from a remote source embodied in a communications medium such as an electronic signal, radio frequency carrier wave or optical carrier wave. Such carrier media are also envisaged as aspects of the present disclosure.

It will be understood by those skilled in the art that, although the present disclosure has been described in relation to the above described example embodiments, the disclosure is not limited thereto and that there are many possible variations and modifications which fall within the scope of the present disclosure.

The scope of the present disclosure includes any novel features or combination of features disclosed herein. The applicant hereby gives notice that new claims may be formulated to such features or combination of features during prosecution of this application or of any such further applications derived therefrom. In particular, with reference to the appended claims, features from dependent claims may be combined with those of the independent claims and features from respective independent claims may be combined in any appropriate manner and not merely in the specific combinations enumerated in the claims.

The invention claimed is:

1. A computer implemented method of access control for a user device having at least one component for determining behaviors of a user, the method comprising:

defining a training period during which authentication of users to access the user device is based on credentials and not on behavior;

during the training period:

obtaining, from the at least one component, data indicating a behavior of the user,

performing credential-based authentication of users to access the user device, and

generating, for each credential-based authentication, an item of training data including at least a result of the credential-based authentication and the behavior of the user, the item of training data being generated both for authenticated users and non-authenticated users, each item of training data generated for an authenticated user including the behavior of the authenticated user and the result of the credential-based authentication being a compliant state and each item of training data generated for a non-authenticated user including the behavior of the non-authenticated user and a result of the credential-based authentication being a non-compliant state;

training a machine learning classifier based on the training data such that the classifier is operable to classify user behavior as compliant or non-compliant such that compliant behavior is determined by the classifier to be consistent with prior behavior for permitting access to the device subsequent to the training period; and

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responsive to a determination that a behavior subsequent to the training period is classified as non-compliant, requesting a credential-based authentication of the user and permitting access to the device in response to the credential-based authentication,

wherein permitting access to the device further includes constructively training the classifier based on the subsequent behavior as a compliant behavior by providing the subsequent behavior as an additional training data example.

2. The method of claim 1 wherein preventing access to the device further includes constructively training the classifier based on the subsequent behavior as a non-compliant behavior by providing the subsequent behavior as an additional training data example.

3. The method of claim 1 wherein the component is one or more of: a location sensor; a position sensor; an orientation sensor; an accelerometer; an input device; a touchscreen; a temperature sensor; a time determiner; a pressure

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sensor; an olfactory sensor; a chemical sensor; a biometric sensor; a heart rate sensor; a cardiogram generator; a sound sensor; a voice recognition component; a handwriting recognition component; a global positioning system; and a gyroscope.

4. The method of claim 1 wherein the credential-based authentication includes one or more of: an authentication scheme using a user identifier and password; a key-based user authentication scheme; a token-based user authentication scheme; and a multi-factor authentication scheme in which authentication is requested via a different device.

5. A computer system including a processor and memory storing computer program code for performing the method of claim 1.

6. A non-transitory computer-readable storage medium comprising computer program code to, when loaded into a computer system and executed thereon, cause the computer system to perform the method of claim 1.

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